Fiscal Year:	FY 2023	**Task Last Updated:**	FY 07/19/2023
PI Name:	Pecaut, Michael Ph.D.		
Project Title:	A Systems-Biology Approach to Assessing the Impact of a Centrifugation Model of Spaceflight on Cross-System Communication		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
PI Email:	mpecaut@llu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8372
Organization Name:	Loma Linda University		
PI Address 1:	Division of Biomedical Engineering Sciences, Department of Basic Sciences		
PI Address 2:	11175 Campus St, Chan Shun Pavilion, A-1016		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	05/14/2019	**End Date:**	05/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/30/2023 per F. Hernandez/ARC (Ed., 5/3/22) NOTE: End date changed to 5/30/2022 per NSSC information (Ed., 9/1/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wilson, Christopher Ph.D. (Loma Linda University) Fleshner, Monika Ph.D. (University of Colorado)		
Grant/Contract No.:	80NSSC19K1038		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Challenges to astronaut health caused by spaceflight-relevant stressors will impact signaling pathways between systems, ultimately leading to degenerative changes as homeostasis is impaired. For example, changes in the microbial environment can result in dramatic changes in various internal and external microbiomes. Both psychological and physiological stress activates the sympathetic nervous system (SNS) and initiates wide-ranging physiologic changes. These changes have serious implications on immune function and the activation of inflammatory pathways. We propose to use centrifugation to characterize the impact of chronic changes in the gravitational environment on cross-system communication.
We hypothesize that hypergravity-induced changes in gut microbiome, SNS function, and inflammatory activity lead to a breakdown in the communication between distinct physiological systems (as measured by changes in cardiopulmonary control) and that these responses depend on the gravitational environment. Furthermore, we propose that these gravity-dependent deficits can be countered by a combination of prebiotic nutritional supplements that are known to preferentially expand stress protective bacterial species.

SPECIFIC AIMS

Specific Aim 1. Confirm that centrifugation results in detectable changes in gut microbiome, as well as in sympathetic and inflammatory activity, indicative of increased health risk. Mice will be placed on the 8 feet diameter centrifuge at Ames Research Center and exposed to 1.5 or 2G for 4 weeks. After centrifugation, adrenal glands, blood, liver, spleen, and intestines will be collected and assessed for changes in sympathetic and inflammatory activity as well as for changes in microbiome as compared to matched 1G controls.

Specific Aim 2. Confirm that prolonged exposure to hypergravity leads to a breakdown in communication between distinct physiological systems, causing decrements in function. The same mice used in Aim 1 will be used here. Our model focuses on brainstem control of cardiopulmonary function, driven primarily by sympathetic/vagal innervation. The brainstem, spinal cord, heart, and lungs will be collected. Analysis will focus specifically on the nucleus tractus solitarii (NTS), the first-order integrative site for vagal afferents projecting from the gut, heart, and lungs to the central nervous system. The NTS coordinates efferent output to targets that are critical for the control of breathing, heart/vasculature, and gut. In vivo and in vitro electrophysiology will be used to quantify changes in sympathetic tone and confirm that synaptic activity within the NTS is altered by changes in microbiome. Immunohistochemistry and RNAseq will be used to assess changes in inflammation and stress-related message, allowing us to determine the role cytokines and downstream effectors play in modulating NTS neuronal activity correlated with cardiorespiratory control.

Specific Aim 3. Determine that centrifugation-induced changes in health status signaling are linked to changes in microbiome. Based on our previous findings, we hypothesize that ingestion of a prebiotic supplement blend [galactooligosaccharides (GOS) + polydextrose (PDX) + lactoferrin + milk fat globule membrane protein (MFGMP)] will both prevent gut microbial dysbiosis produced by centrifugation and increase selective stress protective gut bacteria (i.e., lactobacillus and bifidobacteria). Prebiotic-enriched chow or calorically matched chow will be available ad libitum throughout the experiment, starting 1 week prior to centrifugation. Mice will be placed on the 8 feet diameter centrifuge at Ames Research Center and exposed to 2G for 4 weeks. Parameters described in Aims 1 & 2 will be assessed as described.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Astronauts on long-duration exploratory voyages will be exposed to at least four potentially synergistic risks to astronaut safety: 1] psychological and environmental stress inherent to long-duration space missions; 2] altered inertial forces (i.e., unloading/microgravity); 3] exposure to potentially dangerous levels of microbial contamination; and 4] long-term exposure to low-dose/low-dose-rate radiation. Each of these risks are continually monitored and communicated to the body through a variety of overlapping circulatory and neural mechanisms, ultimately resulting in a profound change in overall physiologic function. In the absence of any external risk to health, physiological systems are continually communicating to maintain homeostasis.
In many ways, these stressors are similar to those encountered in immunocompromised and aging populations on Earth. Like on Earth, challenges to health caused by these stressors will change signaling pathways between systems, ultimately leading to degenerative changes as homeostasis is impaired. For example, 1] Psychological stress is known to activate sympathetic nervous system (SNS) pathways; 2] Unloading (e.g., in bed rest patients) can lead to a reduction in metabolic requirements, thereby altering glucose/lipid metabolism pathways; 3] Changes in the microbial environment (e.g., traveling to a different country or between retirement facilities) can result in dramatic changes in various internal and external microbiomes; and 4] Radiation (e.g., radiotherapy) can cause tissue damage, activating systemic inflammatory pathways. Because astronauts and their animal models do not typically have diseases in flight or during experiments, we can use these conditions to determine how the various stressors impact physiological systems in the absence of disease states.

PROJECT HISTORY & REVIEW OF SPECIFIC AIMS
This project was funded in response to the NASA Research Announcement NNH16ZTT001N-FG, Research Opportunities in Space Biology (ROSBio) - 2016 NASA Omnibus Research Announcement, Appendix G: Solicitation of Proposals for Flight and Ground Space Biology Research. This multidisciplinary project includes three independent investigators and two institutions (Loma Linda University/LLU and the University of Colorado at Boulder/CU Boulder). Internally, we call this experiment, "Spaceflight Impacts the Microbiome-Brain Axis" (SIMBA).

Briefly, challenges to astronaut health caused by spaceflight-relevant stressors can impact signaling pathways between physiological systems. Both psychological and physiological stress are well-known to activate the sympathetic nervous system (SNS), resulting in a wide range of systemic physiologic responses. We proposed using the 8ft diameter centrifuge at the NASA Ames Research Center to characterize the impact of chronic changes in the gravitational environment on cross-system communication. We are focusing on two distinct communication pathways. Dr. Christopher Wilson (also from Loma Linda University) is looking for changes in cardiac and neural electrical activity indicative of disruptions in the communication between the brain stem and the cardiopulmonary system; Dr. Monika Fleshner from the University of Colorado at Boulder is focusing on the Brain-Gut Axis by characterizing changes in inflammatory activity and in the gut microbiome. Furthermore, we propose that gravity-dependent deficits can be countered by a combination of prebiotic nutritional supplements that both protect gut microbial ecology from stress-induced disruption and support/expand health promoting probiotic species.

Task Progress:

SUMMARY OF WORK COMPLETED IN THE LAST 12 MONTHS (6/1/2022 - 05/31/2023)

Impacts of Prebiotic Diet and Altered Gravity on Mouse Gut Microbiome and Home Cage Behavior

Dr. Fleshner and her team at CU Boulder have been analyzing both the behavior and microbiome samples collected during our second run. Challenges to astronaut health caused by spaceflight-relevant stressors impact system-wide signaling pathways, ultimately leading to maladaptive changes and homeostasis impairment. There is evidence from human and rodent studies, for example, that spaceflight and time spent on the International Space Station (ISS) impacts behavior, immune function, autonomic nervous function, and more recently, the gut microbiome. We and others have reported that ingestion of a prebiotic diet preferentially expands stress-protective bacterial species, prevent stress-induced dysbiosis, and increases stress resistance and resilience. We tested the hypothesis that ingestion of a prebiotic diet will reduce the system-wide negative impacts of simulated altered gravity. For this part of the study, infrared cameras allowed 24h video recordings of home cage behavior at 3G and 1G. Behavioral Observation Research Interactive Software (BORIS) was used to score home cage recordings. Fecal samples were collected before centrifugation and once a week throughout the run. Fecal samples were sequenced in multiplex on an Illumina HiSeq 2000 and gene sequence processing was done using QIIME2. Amplicon sequence variants (ASVs) were generated using the deblur algorithm, and taxonomy classification was done via the QIIME2 feature classifier plugin and based on Silva.

Briefly, exposure to 3G reduced food consumption and body weight, had a greater impact on behavior during the inactive cycle (light) than the active cycle (dark), and altered the fecal microbial ecology (beta diversity). Specifically, 3G increased Clostridium Sensu Stricto.

The prebiotic diet prevented changes in home cage behavior and prevented the changes noted in the microbiome produced by 3G. Additionally, we found that the prebiotic diet increased S. Dubosiella newyorkensis, a recently characterized probiotic species.

Overall, prebiotic diet increased stress-protective bacterial species, prevented stress-induced dysbiosis, and reduced the system-wide negative impacts of simulated altered gravity.

SIMBA Heart Rate Variability (HRV) Analysis

Dr. Chris Wilson at LLU focused on the heart rate analysis. Heart rate variability (HRV) is a useful tool for characterizing changes in the autonomic nervous system, meaning both sympathetic and parasympathetic activity, from electrocardiogram (ECG) recordings. In this experiment, we hypothesized that increased sympathetic activity (the “fight or flight” response) would occur when mice were exposed to high gravity (3 times normal terrestrial gravity, or 3G). We anticipated that a prebiotic diet, a diet designed to stabilize gut health and provide an anti-inflammatory state in the gut, would protect against stress responses at 3G. The communication between the gut and brain is known to be important in reducing chronic inflammatory status and maintaining normal gut health. Due to complications with the vivarium groups, we only compared groups housed in the centrifuge room for this analysis.

Mice were placed on the prebiotic diet at least two weeks prior to being exposed to 3G. Even before centrifugation started (Week 0), the mean heart rate was lower in the prebiotic fed groups. However, this only reached significance in the prebiotic 3G (PB-3G) group when compared to the control diet groups (CD-1G & CD-3G) in post hoc analysis. After two weeks of centrifugation, the mice eating the control diet, regardless of the presence of hyper-gravity, had higher mean heart rates, suggesting that their overall sympathetic tone was increased compared to the prebiotic diet treated mice. By Week 4, there were no significant differences between groups. Interestingly, the PB3G group had no significant difference in mean heart rate (HR) data across all weeks compared to 1G controls, suggesting that the prebiotic diet reduced the heart rate response to centrifugation.

In addition to heart rate, our Python-based Jupyter notebook included multiple metrics that are commonly used in quantifying heart rate variability [Task Force, 1996] from ECG, including time-series measurements (SDNN, RMSSD, SDSD, NN50) and frequency domain measurements (SD1, SD2, SD1/SD2 ratio, Ellipse Area, VLF, LF, HF, normalized LF, and HF, and LF/HF ratio), as well information theory metrics (Shannon, Approximate, and Sample entropy). These metrics are defined below. [Ed. Note: See "Definitions" section below.]

Frequency based HRV measurements

The first four of these (SD1, SD2, SD1/SD2 ratio and Ellipse Area) are all metrics based on Poincaré scatter plots and provide both an initial qualitative and second-order quantitative measure of heart rate variability and adaptability—the more variability (less rigidly “clock” like), the healthier the heart. Too much variability indicates an unhealthy heart and too little variability indicates damage to the heart, respectively. So HRV is a biological representation of the “Goldilocks” phenomenon, cardia variability has to be “just right” to respond appropriately to stress. For example, a tight cloud in the Poincaré plot (i.e., Ellipse Area) indicates low variability and high-sympathetic drive (more stress). Larger clouds and greater ellipse area indicates a higher level of healthy variability (more parasympathetic drive). In these experiments, there was an increase in overall Ellipse Area for the 3G groups (both control and prebiotic) with the greatest increase in Ellipse Area seen in the prebiotic, 3G group. This suggests that the prebiotic diet provides an increase in HRV remaining in the “just right,” more adaptable, range of activity.

Spectral measures of HRV (LF, HF, and LF/HF ratio) are believed to represent sympathetic nervous system activity, parasympathetic nervous system activity, and the relative ratio of these two as an overall index of autonomic tone in driving cardiac rhythm. Not surprisingly, median changes in LF/HF ratio were greatest in the second week of the protocol for the control-diet 3G group, suggesting a higher overall sympathetic drive to the heart. However, the prebiotic 3G group, while having a wider interquartile range of LF/HF, had a median value that was the same as pre-spin and for week 4, suggesting that the cardiovascular system was more resilient to stress than the control diet group at 3G.

Time-series based HRV measurements

SDNN (standard deviation of all N-N intervals) is a measure of the variability in the time between heart beats (N-N interval) and a “general” metric of overall HRV. In humans, if this measure is too low, the heart is generally considered unhealthy and at risk for heart disease. Similarly, SDSD (standard deviation of differences between N-N intervals) is the variability in the differences between beat-to-beat intervals. This measure is more related to short-term variability in beats. Lower SDSD is indicative of poorer heart health and increased stress. RMSSD (square root of the mean sum of squares of the differences between adjacent N-N intervals) is the beat-to-beat variability of heart rate and is linked to changes in vagus nerve activity (the gut-brain link is via the vagus nerve), particularly in the ability of the heart to

	respond to stress. Lower values imply less parasympathetic tone and a reduced capacity to adaptively respond to stress. SDNN, SDSD, and RMSSD were significantly higher in the prebiotic treated groups during Week 2, suggesting that these mice had a more resilient cardiovascular system. So, the mice fed the prebiotic diet are both actively adapting to stress, and are facile at maintaining normal physiological cardiovascular state. Overall, these data suggest that by Week 2 of centrifugation, the mice still had increased levels of vagal activity/autonomic tone. By Week 4, the centrifuged mice showed signs of a re-balancing of autonomic tone and adaptation to hypergravity. This adaptation was particularly pronounced in the prebiotic treated mice, suggesting that a prebiotic, anti-inflammatory diet has promise to reduce overall cardiovascular stress and improve physiological adaptability. This is encouraging when one considers the importance of adaptive responses to stress in altered gravity. A preliminary hypothesis based on this data could be that astronauts subjected to prolonged changes in the gravitational environment (> 24 hours) would be able to adapt to the increased gravitational field and show no debilitating cardiovascular stress. Ideally, further work focused on changes in autonomic control of the cardiovascular system in response to a prebiotic diet would be an obvious and, potentially, fruitful line of investigation to provide a pathway toward an optimized diet for space travelers. DEFINITIONS SD1 - The width of the Poincaré ellipse. Provides an index of distributional variability. SD2 - The length of the Poincaré ellipse. SD1/SD2 Ratio - The ratio of SD1/SD2 that measures the inter-event variability in a time series. Ellipse Area - The area of the Poincaré ellipse. HF - High frequency (HF) band (1.5–4.0 Hz in mice). Correlated with parasympathetic activity. LF - Low frequency (LF) band (0.4–1.5 Hz in mice). Correlated with sympathetic activity. LF/HF Ratio - Ratio of LF to HF power. Used as an index of sympathetic-to-parasympathetic tone. Shannon Entropy - Measure of variability. Quantifies the amount of information in the variable (in this case, beat-to-beat interval). Sample Entropy - Measure of regularity/complexity of continuous physiological time-series information. SDNN - Standard deviation of the interbeat intervals (IBI) of normal sinus beats. Simplest overall metric of HRV. SDSD - Standard deviation of the successive differences between adjacent R-R intervals. RMSSD - Time Square root of successive NN intervals. Another common metric of HRV.
Bibliography Type:	Description: (Last Updated: 08/04/2025)
Articles in Peer-reviewed Journals	Iwakoshi N, Kelley T, Hopkins S, Thompson R, Tan J, Pecaut M, Fleshner M, Wilson C. "A prebiotic diet increases cardiovascular adaptability under hypergravity-induced stress." Physiology. 2024 May 21;39(S1):1751. https://doi.org/10.1152/physiol.2024.39.S1.1751 , May-2024
Articles in Peer-reviewed Journals	Thompson RS, Hopkins S, Kelley T, Wilson CG, Pecaut MJ, Fleshner M. "A prebiotic diet containing galactooligosaccharides and polydextrose attenuates hypergravity-induced disruptions to the microbiome in female mice." Nutrients. 2025 Jul 24;17(15):2417. https://doi.org/10.3390/nu17152417 , Jul-2025